Translating Research Gaps into Method

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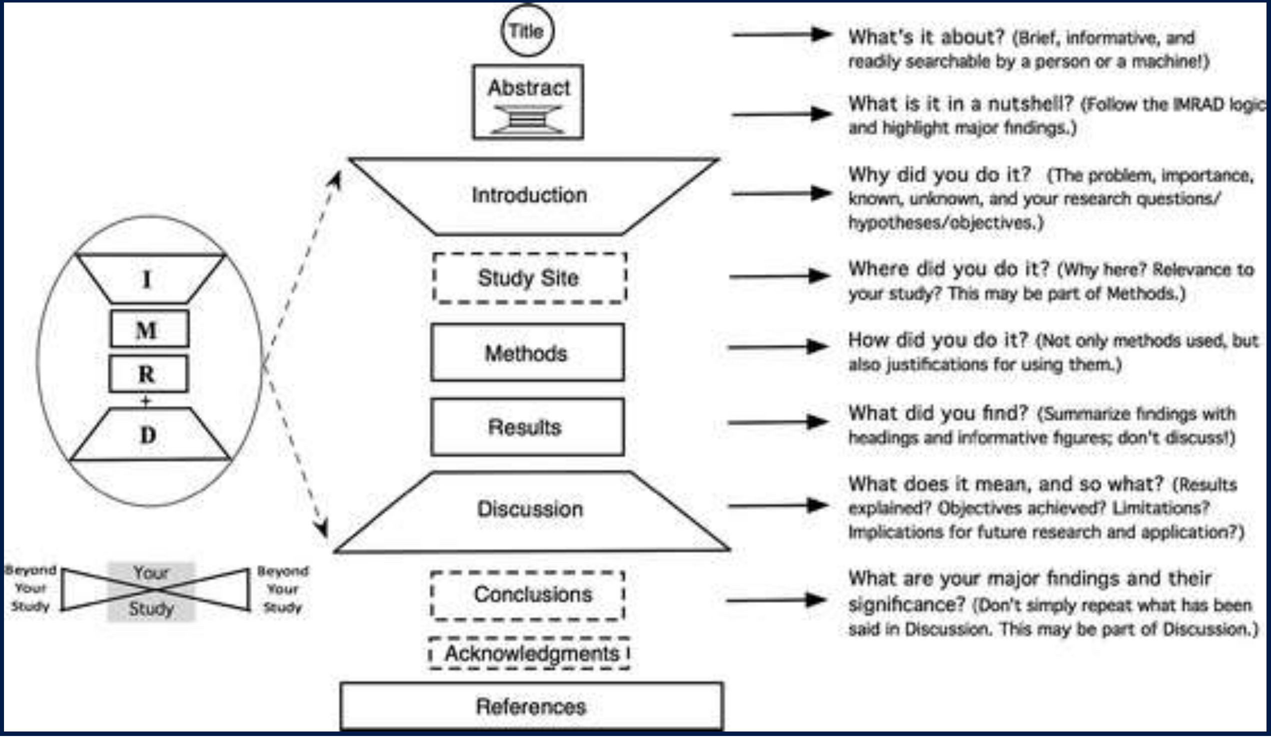
Outline

- Identify and articulate research gaps
- Translate gaps into research objectives, questions and hypothesis
- Align objectives with quantitative and social science methods

Recap on IMRAD method



Recap on Abstract

- General background
- Specific background
- Knowledge gap
- Here we show...
- Results
- Implications

Recap on Abstract

ABSTRACT

Maize Lethal Necrosis (MLN) is a viral maize disease that suddenly appeared in Kenya in the last few years and is causing major damages. Interventions are urgently needed, but information is lacking on its geographic distribution and losses caused. Group discussions were held in 2013 in 121 sublocations in the major maize zones, and respondents were asked if they had heard about MLN, when they first observed it, the proportion of households affected, and the estimated yield loss in affected areas. Responses were used to estimate the proportion of maize lost in the community, and these results were interpolated and combined with maize production data to estimate quantities lost. Western Kenya suffered most with more than half of farmers affected, followed by Central and Eastern Kenya, with up to a third of farmers. Yield losses in affected areas were highest in Western Kenya, followed by the highlands (in Central Kenya and the Rift Valley) and at the coast, but were low in the drylands (in Eastern Kenya). Total maize losses in Kenya were estimated at 0.5 million ton per year, or 22% of the average annual production before MLN, with a value estimated at $180 million. Losses were concentrated in Western Kenya, in particular the moist transitional zone (58% of all maize lost) and the moist mid-altitudes (19%), but also in the highlands (17%). Losses were small in the drylands and at the coast. Urgent action is needed to help farmers cope. In the short term, they need to be informed about the disease and provided with advice on appropriate agronomic practices. In the long term, varieties resistant to MLN need to be developed, first for the moist transitional and moist mid-altitude zones and followed by the highlands, but ultimately for all agroecological zones.

Recap on Abstract

Mental health treatment in Kenya: task-sharing challenges and opportunities among informal health providers

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Abstract

Background: The study was conducted to explore challenges faced by trained informal health providers referring individuals with suspected mental disorders for treatment, and potential opportunities to counter these challenges.

Methods: The study used a qualitative focus group approach. It involved community health workers, traditional and faith healers from Makueni County in Kenya. Ten Focus Group Discussions were conducted in the local language, recorded and transcribed verbatim and translated. Using a thematic analysis approach, data were entered into NVivo 7 for analysis and coding.

Results: Results demonstrate that during the initial intake phase, challenges included patients' mistrust of informal health providers and cultural misunderstanding and stigma related to mental illness. Between initial intake and treatment, challenges related to resource barriers, resistance to treatment and limitations of the referral system. Treatment infrastructure issues were reported during the treatment phase. Various suggestions for solving these challenges were made at each phase.

Conclusions: These findings illustrate the commitment of informal health providers who have limited training to a task-sharing model under difficult situations to increase patients' access to mental health services and quality care. With the identified opportunities, the expansion of this type of research has promising implications for rural communities.

Keywords: Informal health providers, Task sharing, Mental health, Challenges, Kenya

Discuss some examples from participants

What Is a Research Gap? Gap-Spotting vs. Problematization

Gap-Spotting

Identifying something existing literature has not yet covered, the most common approach

Problematization

Questioning the underlying assumptions of existing literature itself, rarer, often more influential

Common gap types

- **Empirical** → not yet measured in a setting
- **Theoretical** → existing theory doesn't explain it
- **Methodological** → prior methods had limitations
- **Contextual/geographic** → untested in this region
- **Population** → an under-studied group

Moving from Gap to Objectives: Aims, Questions, Hypotheses

Term	Function	Examples
Aim	Broad overall purpose	• Improve understanding of smallholder drought resilience • Assess the impact of urban greening on urban quality of life
Objective(s)	Specific, achievable steps	• Quantify yield effect of a drought-tolerant variety under three rainfall scenarios • Develop a model for predicting the influence of climate on mental health
Research question	Open question (qualitative/exploratory)	• How do farmers decide whether to adopt a new variety? • Does increase in diurnal temperature influence sleep patterns of children?
Hypothesis	Testable, falsifiable prediction (quantitative)	H1: Yield loss under drought is lower for the new variety

Reread your abstract and identify a research aim, objective or question

Aligning Objectives to Methods: The Golden Thread

Gap → Objective → Question/Hypothesis → Method → Analysis → Discussion

Measure / quantify / compare / test a relationship	→ Quantitative methods (experiments, surveys with statistics, modelling)
Explore / understand / explain a process or meaning	→ Qualitative methods (interviews, focus groups, case studies, ethnography)
Requires both magnitude and meaning	→ Mixed methods

Quantitative Example: Agricultural Field Trial

IMRAD Element	Content
Gap	Limited independent quantification of yield differences between a drought-tolerant variety and the local check
Objective	Quantify the yield difference across a gradient of drought stress
Hypothesis	H1: Mean yield loss under drought is significantly lower for the drought-tolerant variety
Methods	Randomized complete block field trial; ANOVA and regression against a drought stress index
Discussion	Whether the yield advantage justifies variety replacement programmes

Qualitative Example: Perceptions of Early Warning Systems

IMRAD Element	Content
Gap	Unclear how pastoralist communities perceive and trust a new drought early-warning system
Research question	How do community members perceive the credibility and usefulness of the system?
Methods	Case study design: focus group discussions and key informant interviews; framework analysis
Discussion	Implications for redesigning the system's communication and trust-building

A Mixed-Methods Example

Convergent

Quantitative + Qualitative strands collected in parallel, merged at interpretation to see where they agree or diverge

Explanatory Sequential

Quantitative phase (e.g., survey) followed by qualitative phase (e.g., interviews) to explain the results

Exploratory Sequential

Qualitative phase identifies locally meaningful strategies; informs a subsequent quantitative instrument

Example: Quantify how widespread a practice change (e.g. indigenous seed propagation) has been (survey) → Explain why (interviews with a subsample) → integrate both in the Discussion.

Writing the Methods Section to Match Objectives

- Does every objective have a corresponding method that could plausibly answer it?
- Is the sampling strategy appropriate and justified for that objective?
- Is the data collection procedure described in enough detail for replication?
- Is the analysis plan explicitly linked to the hypothesis or question it addresses?
- Are ethical considerations addressed?

- Write according to the specifications of your field
- Explain your methodological and data collection approaches in detail
- Explain, each methodological step in detail so that others can replicate your work ← Transparency
- When apply other’s methods and formulars , properly cite
- Ensure that each objective has been adequately addressed in the methods

Examples

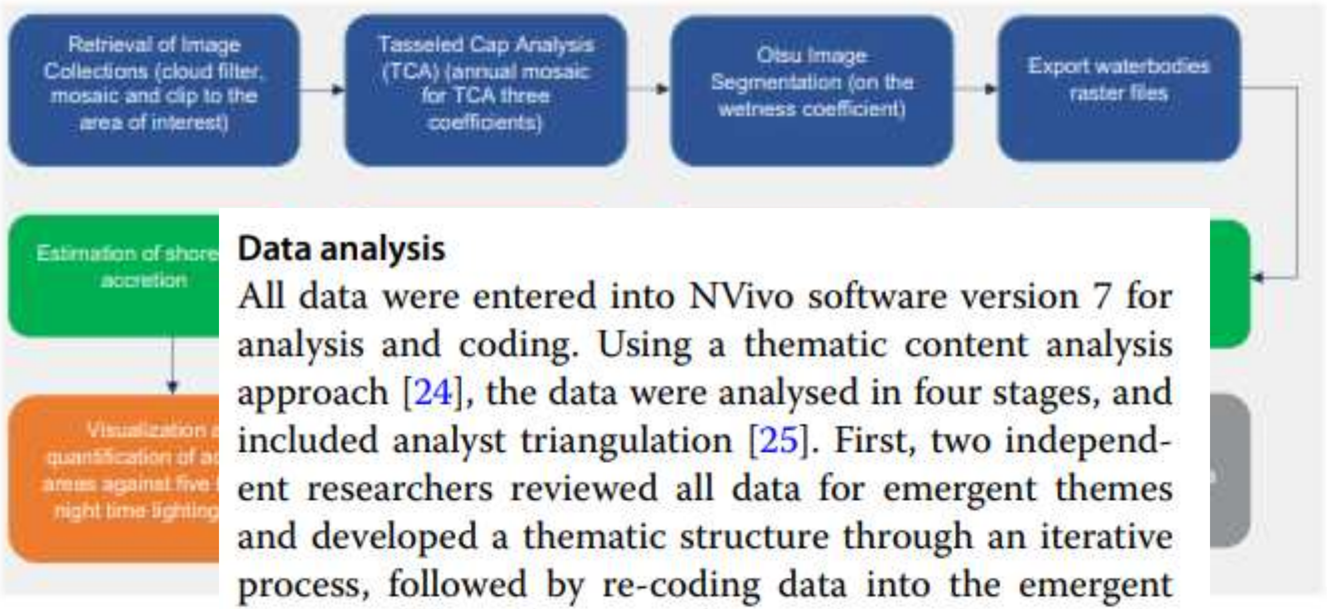
2.1. Data Collection

Three main data sources were used in this work including, official road networks from national mapping agency in Kenya, roads and tracks from OSM, and GPS tracks and points that were collected as part of the motorcycle tracking experiment.

At the time of the experiment, there were approximately 400 motorcycle taxi riders in and around Sigomre market. Commonly, riders organize themselves into groups of about 100 riders who make up a self-help group. In Sigomre market, there are four groups. Ten (10) volunteer riders were selected from market center to participate in the experiment. The volunteer riders were tagged on their wrists and tracked on a daily basis from 6 a.m.–9 p.m. for a duration of two weeks. At the end of each week, data loggers were retrieved from the riders and the data recorded by each logger within the week downloaded and archived. Additionally, in each day of the week, we interacted with the group of volunteer riders to confirm that the data loggers were adequately charged and in good working condition. **Figure 2** represents, (a) typical village tracks that were traversed by the motor cycle taxis, (b) the particular logger that we used in this experiment together with the wrist strap and the charging and data download cable, and (c) an example of a volunteer riding a motorcycle taxi while recording GPS locations through a GPS data logger strapped on his wrist.

Figure 2. Illustration of (**a**) Typical rural motorcycle tracks (**b**) i-gotU GT-600 Global Positioning System (GPS) travel and sports data logger (**c**) A motor cycle rider strapping a GPS data logger on the wrist while riding a motorcycle taxi.

A detailed description of data collection methods



Data analysis

All data were entered into NVivo software version 7 for analysis and coding. Using a thematic content analysis approach [24], the data were analysed in four stages, and included analyst triangulation [25]. First, two independent researchers reviewed all data for emergent themes and developed a thematic structure through an iterative process, followed by re-coding data into the emergent themes. All coding was discussed, any discrepancies resolved, and full agreement reached.

Figure 2. Flow d … n human influence. The st … GEE. The steps highlighted in grey outline the preparation of the nighttime light data in GEE. The steps outlined in green (the analysis of shoreline dynamics) and orange (the analysis of nighttime light data) were conducted in a standard desktop GIS platform.

Using graphical aides to support in documenting methods

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Description of qualitative data analysis methods and citing additional methods

Assignment: Relook at your work, identify the gap, research objectives, and potential methods for each objective, and organize as below

IMRAD Element	Content
Gap	State a clear gap of your research (1)
Research Aim	State the overall aim/purpose of your research
Specific Objectives	Outline 3-4 specific objectives of your research
Methods	For each specific objective, outline and briefly describe the main methodological steps

Questions and discussions